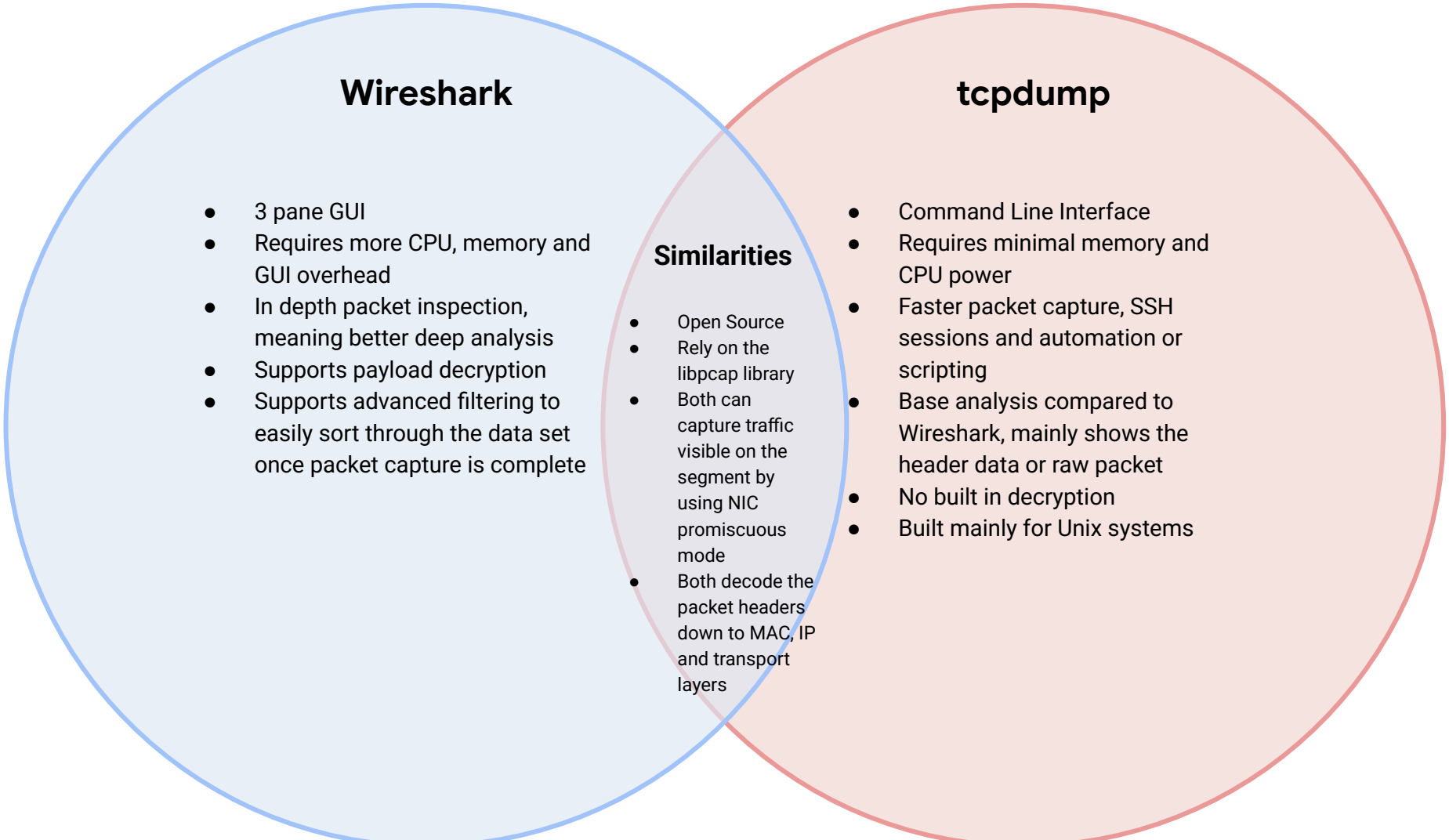
Wireshark
3 pane GUI
Requires more CPU, memory and GUI overhead
In depth packet inspection, meaning better deep analysis
Supports payload decryption
Supports advanced filtering to easily sort through the data set once packet capture is complete
Similarities
Open Source
Rely on the libpcap library
Both can capture traffic visible on the segment by using NIC promiscuous mode
Both decode the packet headers down to MAC, IP and transport layers
tcpdump
Command Line Interface
Requires minimal memory and CPU power
Faster packet capture, SSH sessions and automation or scripting
Base analysis compared to Wireshark, mainly shows the header data or raw packet
No built in decryption
Built mainly for Unix systems